

From Electronic Structure to Continuum Response: A Computational Model for Strain-Dependent Conductivity in Piezoresistive Materials

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Abstract

Many sensing devices rely on transduction mechanisms that convert mechanical deformation into measurable electrical signals. A prominent example is the piezoresistive effect, where the electrical conductivity of a material varies under applied strain.

From a materials physics perspective, this behavior originates from strain-induced modifications of the electronic structure and carrier transport properties. Mechanical deformation alters band energies, effective masses, and scattering mechanisms, which in turn influence carrier mobility and electrical conductivity [3]. In certain regimes, additional effects such as tunneling may also contribute to the overall response.

This work develops a computational framework for modeling strain-dependent conductivity in deformable materials by connecting these physical mechanisms across scales. At the continuum level, the mechanical behavior is described by the equations of elasticity, while the electrical response is governed by a conductivity equation with strain-dependent coefficients. This leads to a coupled system of partial differential equations of the form

$$\nabla \cdot \boldsymbol{\sigma}(u) = 0, \quad \nabla \cdot (\sigma(\varepsilon(u)) \nabla V) = 0,$$

where u denotes the displacement field, V the electrical potential, and $\sigma(\varepsilon(u))$ represents an effective conductivity incorporating the influence of mechanical deformation.

A central aspect of this study is the interpretation of the constitutive relation $\sigma(\varepsilon)$. Rather than treating it purely phenomenologically, the framework is designed so that it can be informed by transport models derived from electronic structure calculations, providing a physically motivated link between microscopic behavior and macroscopic response.

From a computational perspective, the focus lies on the formulation and numerical solution of the coupled system using finite element methods. A baseline case with constant conductivity is first analyzed and validated, followed by the extension to spatially varying, strain-dependent conductivity. The framework is constructed to remain flexible, allowing the incorporation of physics-based models or data-driven approximations for $\sigma(\varepsilon)$.

The proposed approach provides a consistent and extensible pathway for studying electromechanical coupling in materials. Potential applications include sensing technologies, flexible electronics, and structural health monitoring.

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1 Introduction

The piezoresistive effect arises from the coupling between mechanical deformation and electrical conduction. In typical applications, a doped semiconductor element (such as silicon) is embedded within a mechanical structure, where external forces induce strain in the material. This deformation alters the electronic transport properties of the material, leading to measurable changes in its electrical response, commonly observed as variations in voltage or resistance.

The piezoresistive effect, first observed by William Thomson (Lord Kelvin) in the 19th century and later developed for semiconductors by Bardeen, Shockley, and Smith [6], describes the dependence of electrical conductivity on mechanical deformation. At a microscopic level, this behavior

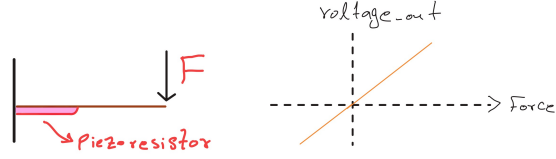


Figure 1: A schematic representation of the piezoresistive effect, where mechanical deformation leads to changes in electrical response.

originates from strain-induced modifications of the electronic structure and carrier transport mechanisms. In particular, quantities such as band energies, effective masses, and scattering processes are affected by deformation, which in turn influence carrier mobility and conductivity [3].

There exists a wide spectrum of engineering realizations of the piezoresistive effect, including strain gauges, pressure sensors, microelectromechanical systems (MEMS), cantilever-based sensing devices, and flexible or wearable systems [1]. Although these systems differ in geometry, material composition, and boundary conditions, they can be interpreted as different realizations of the same underlying electromechanical coupling.

Despite extensive technological development and optimization, the modeling of this coupling is often carried out in a phenomenological and application-specific manner, where the dependence of conductivity on strain is introduced empirically. From a computational materials perspective, this raises a more fundamental question: how can such macroscopic descriptions be related in a consistent way to the underlying physical mechanisms governing electronic transport?

The present work approaches this problem by developing a computational framework in which the electromechanical coupling is described at the continuum level, while maintaining a clear connection to the underlying transport physics. In this setting, the continuum model serves as an effective description of strain-dependent conductivity, acting as a bridge between microscopic electronic behavior and macroscopic response.

2 Phenomena of piezoresistivity

3 Continuum Model Formulation

4 Numerical Solution of the continuum problem

5 Essentials of Strain and Stress Analysis

6 Symmetry and operations analysis of the crystalline Si

7 Deformation Potentials

8 Results and discussions

9 Error analysis

10 Problem Articulation

At the continuum level, electrical conduction in a deformable material can be described by a conductivity equation in which the electrical conductivity depends on the local mechanical strain. Such models are widely used in engineering applications and can be solved efficiently using numerical methods. However, the central quantity in this description, namely the strain-dependent conductivity $\sigma(\varepsilon)$, is often introduced phenomenologically, without an explicit connection to the underlying material physics.

From a materials physics perspective, this raises a fundamental question: how does mechanical deformation at the atomic scale translate into a change in macroscopic electrical conductivity?

In semiconductors, strain modifies the crystal structure and therefore the electronic band structure of the material. This leads to changes in quantities such as band energies, effective masses, and carrier velocities. As a consequence, the transport properties of charge carriers are altered, affecting mobility and electrical conductivity. These mechanisms are described in semiconductor theory through band-structure analysis and transport models such as the semiclassical Boltzmann transport equation [3].

This observation suggests a natural multiscale structure of the problem. At the microscopic level, strain influences the electronic structure of the material. At the transport level, the modified electronic structure determines carrier dynamics and effective conductivity. At the continuum level, this conductivity enters as a constitutive parameter in the electrical conduction equation.

From a transport perspective, the electrical conductivity can be expressed using the semiclassical Boltzmann transport equation as

$$\sigma_{ij} = \frac{e^2}{4\pi^3} \sum_n \int v_i^n(\mathbf{k}) v_j^n(\mathbf{k}) \tau_n(\mathbf{k}) \left(-\frac{\partial f_0}{\partial E} \right) d\mathbf{k}, \quad (1)$$

where $v_i^n(\mathbf{k})$ denotes the group velocity of electrons in band n , τ_n is the relaxation time, and f_0 is the equilibrium distribution function.

Mechanical deformation enters this description through strain-induced shifts in the electronic band structure, which can be modeled using deformation potential theory,

$$\delta E_n = \Xi_{ij} \varepsilon_{ij}, \quad (2)$$

where Ξ_{ij} denotes the deformation potential tensor.

Together, these relations provide a microscopic interpretation of the strain-dependent conductivity $\sigma(\varepsilon)$ used in the continuum model.

The core challenge of this work is therefore to connect these levels in a computationally consistent way. Rather than treating $\sigma(\varepsilon)$ purely as a fitting function, the goal is to interpret it as an effective quantity that reflects underlying electronic and transport mechanisms, while remaining suitable for use in continuum simulations.

This perspective allows the continuum model to serve not only as a numerical tool, but also as a bridge between microscopic physical processes and macroscopic device behavior.

The multiscale structure of the problem can be summarized as follows:

Scale	Description	Typical Tool
Electronic	Band structure $E_n(\mathbf{k}, \varepsilon)$	DFT
Transport	Conductivity $\sigma(\varepsilon)$	Boltzmann transport
Continuum	Fields $V, \mathbf{E}, \mathbf{J}$	FEM

In particular, this framework enables the incorporation of physically informed or data-driven constitutive relations, depending on the level of detail available for the underlying material.

11 Problem Statement and Motivation

Consider a conductive element with length L , cross-sectional area A , and resistivity ρ . Its electrical resistance is given by

$$R = \frac{\rho L}{A}. \quad (3)$$

Under mechanical loading, both the geometry and the intrinsic material properties of the conductor may change. In particular, the length, cross-sectional area, and resistivity become functions of deformation, making the resistance dependent on the mechanical state.

Using Ohm's law,

$$R = \frac{V}{I}, \quad (4)$$

and expressing voltage and current in terms of field quantities,

$$V = EL, \quad I = JA, \quad (5)$$

we obtain

$$\frac{EL}{JA} = \frac{\rho L}{A}. \quad (6)$$

Canceling the geometric factors L and A leads to the local relation

$$\mathbf{J} = \sigma \mathbf{E}, \quad \sigma = \frac{1}{\rho}, \quad (7)$$

where $\mathbf{E} = -\nabla V$ is the electric field. This formulation expresses Ohm's law in a local (continuum) form, independent of the specific geometry and applicable to spatially varying fields.

Combining this with charge conservation in the steady state,

$$\nabla \cdot \mathbf{J} = 0, \quad (8)$$

yields the conductivity equation

$$\nabla \cdot (\sigma \nabla V) = 0. \quad (9)$$

This equation provides a continuum description of electrical conduction in materials, where the conductivity σ may vary spatially. It belongs to the class of elliptic partial differential equations in divergence form [2].

In piezoresistive materials, the key feature is that the conductivity is no longer constant, but depends on the mechanical deformation of the material. At a macroscopic level, this dependence can be expressed as

$$\sigma = \sigma(\varepsilon(u)), \quad (10)$$

where the strain tensor is given by¹

$$\varepsilon(u) = \frac{1}{2}(\nabla u + \nabla u^T). \quad (11)$$

This relation can be interpreted as an effective description of how mechanical deformation influences electrical transport. From a physical perspective, the dependence of conductivity on strain originates from strain-induced modifications of the electronic structure and carrier transport properties, which are affected by lattice deformation [3].

The resulting model therefore introduces a nonlinear coupling between mechanical deformation and electrical conduction: the mechanical field u determines the strain $\varepsilon(u)$, which modifies the conductivity σ , and this in turn affects the electrical potential V through the conductivity equation.

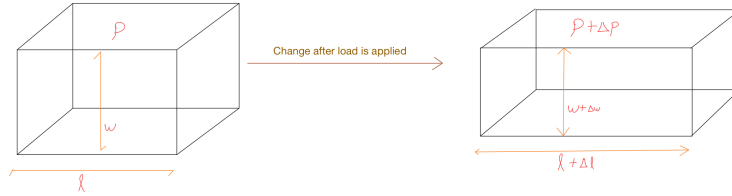


Figure 2: Change of geometric and material properties under mechanical deformation. L denotes the length of the conductor, W its characteristic width, and $A = W^2$ the cross-sectional area (assuming a square cross-section). The material property ρ represents the electrical resistivity, which may vary under mechanical deformation. Both geometric changes and intrinsic material variations contribute to the overall change in resistance.

This illustrates that both geometric deformation and intrinsic material changes contribute to the variation of resistance, providing the basis for modeling the piezoresistive effect within a continuum framework.

¹The strain tensor is defined as the symmetric part of the displacement gradient, which removes rigid body rotations and captures the true deformation of the material.

12 Mathematical Model

Based on the physical considerations discussed above, the electromechanical behavior of the material is described at the continuum level by a coupled system of partial differential equations.

The mechanical response is governed by the equations of elasticity,

$$\nabla \cdot \boldsymbol{\sigma}(u) = 0, \quad (12)$$

where u denotes the displacement field and $\boldsymbol{\sigma}(u)$ is the stress tensor, which depends on the strain through a constitutive relation.

The electrical response is described by the conductivity equation,

$$\nabla \cdot (\sigma(\varepsilon(u)) \nabla V) = 0, \quad (13)$$

where V is the electrical potential and the conductivity σ depends on the strain tensor $\varepsilon(u)$.

Together, these equations define a coupled system in which the mechanical field u determines the strain $\varepsilon(u)$, which in turn modifies the conductivity σ , and thereby influences the electrical potential V .

From a physical perspective, this coupling provides an effective continuum description of how mechanical deformation alters the underlying electronic transport properties of the material.

The system is nonlinear due to the dependence of the conductivity on the strain field.

This formulation serves as the basis for the computational modeling of the piezoresistive effect, linking continuum mechanics with strain-dependent electrical conduction.

13 State of Research

The piezoresistive effect has been extensively investigated in both experimental and engineering contexts, particularly in semiconductor-based sensors and microelectromechanical systems (MEMS) [6, 1]. Early studies established the strong dependence of electrical conductivity on mechanical strain in materials such as silicon, highlighting the importance of crystal structure and carrier transport in determining the sensitivity of the effect.

From a materials physics perspective, piezoresistivity originates from strain-induced modifications of the electronic structure and charge transport properties. Mechanical deformation can alter band structure, effective masses, and scattering mechanisms, which in turn influence carrier mobility and electrical conductivity [3]. These effects are well understood at the microscopic and transport levels.

In practical modeling and engineering applications, however, this coupling is typically introduced in a simplified manner, where the conductivity is treated as an empirical function of strain. While such approaches are effective for device design, they do not provide a transparent link between microscopic transport mechanisms and macroscopic behavior.

At the continuum level, partial differential equation models are widely used to describe both elasticity and electrical conduction. Nevertheless, the incorporation of strain-dependent transport properties into these models is usually carried out phenomenologically, without a systematic connection to the underlying physics.

This reveals a gap between physically grounded transport descriptions and continuum-level modeling. Bridging this gap in a computationally consistent way remains an open and relevant problem, particularly for materials and devices where electromechanical coupling plays a central role.

Based on these considerations, the proposed study is guided by the following research questions:

- How can strain-dependent conductivity be incorporated into a consistent continuum model of electromechanical coupling?
- How does mechanical deformation influence the spatial distribution of electrical potential in conductive materials?
- What effects arise from the nonlinear coupling between mechanics and electrical transport?
- To what extent can flexible (e.g. data-driven or physics-informed) approaches improve the modeling of $\sigma(\varepsilon)$?

To address these questions, the study is structured around the following components:

- Physical background and modeling assumptions
- Formulation of the coupled electromechanical model
- Analysis of the nonlinear system
- Computational methods and numerical experiments
- Interpretation of results in the context of material behavior

14 Methodology

The proposed study follows a computational approach based on a continuum description of the coupled electromechanical system. The methodology is structured in several stages, combining physical modeling with numerical approximation.

First, the governing equations of elasticity and electrical conduction are formulated within a consistent continuum framework. The mechanical problem provides the displacement field u , from which the strain tensor $\varepsilon(u)$ is computed. This strain field enters the electrical problem through the strain-dependent conductivity $\sigma(\varepsilon)$.

As a baseline, the electrical conduction problem with constant conductivity is considered and solved numerically. This case admits an analytical solution and serves to validate the numerical implementation. A finite element discretization is employed, and the numerical solution is compared against the exact solution to ensure correctness.

This baseline case also provides a reference for interpreting deviations introduced by strain-dependent conductivity in the coupled problem.

Building on this, the full coupled system is addressed by introducing a spatially varying conductivity $\sigma(\varepsilon(u))$. The problem is solved iteratively: starting from an initial displacement field, the strain is computed, the conductivity is updated, and the electrical problem is solved. This procedure is repeated until convergence is achieved.

The resulting system is nonlinear due to the dependence of σ on $\varepsilon(u)$, and particular attention is given to the stability and convergence of the numerical scheme, as well as to the sensitivity of the solution with respect to deformation.

From a computational perspective, the finite element method (FEM) is used as the primary discretization technique due to its flexibility in handling heterogeneous material properties and complex geometries. The implementation emphasizes clarity and modularity, allowing systematic exploration of different forms of the strain–conductivity relation.

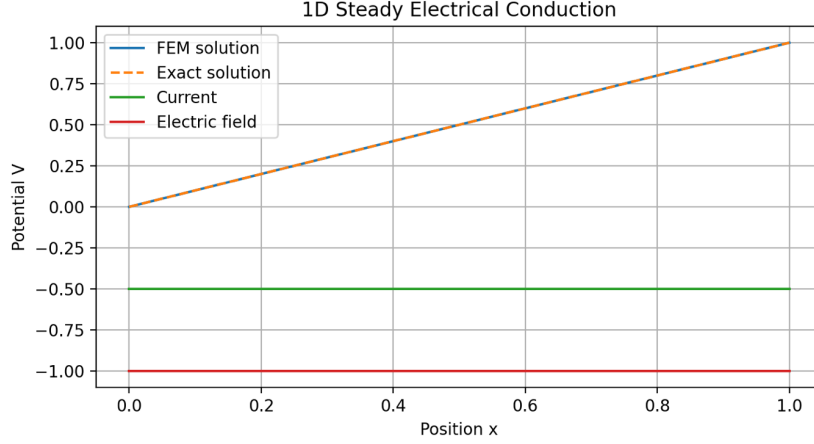


Figure 3: Finite element solution of the 1D steady conduction problem with constant conductivity compared with the analytical solution. The linear potential profile and constant electric field confirm the expected physical behavior.

Beyond purely phenomenological models, the framework is designed to accommodate physically motivated constitutive relations. In particular, the function $\sigma(\varepsilon)$ can be informed by transport models derived from electronic structure calculations, or approximated using data-driven approaches when such data is available.

Overall, the methodology provides a transparent and extensible computational pipeline that connects continuum modeling with physically grounded descriptions of strain-dependent conductivity.

15 Parametric Solution

In modeling electromechanical transport, it is important to distinguish between *geometric effects* and *material effects*. Geometric changes arise from deformation of the domain (e.g., changes in length L or cross-sectional area A), which influence quantities such as resistance through the relation

$$R = \frac{L}{A\sigma}. \quad (14)$$

Material effects, in contrast, describe how intrinsic properties, such as the electrical conductivity σ , change due to deformation. In semiconductors, this latter effect—known as *piezoresistivity*—is typically dominant and is the focus of the present work.

To incorporate this dependence, the conductivity is treated as a function of the mechanical strain,

$$\sigma = \sigma(\varepsilon), \quad (15)$$

where ε denotes a scalar strain measure derived from the strain tensor $\varepsilon(\mathbf{u})$ introduced in Section 12.² The exact functional form of $\sigma(\varepsilon)$ is generally unknown or analytically intractable when derived from first-principles microscopic transport theory. For sufficiently small deformations, however, $\sigma(\varepsilon)$ may be assumed smooth and approximated by its Taylor expansion about the undeformed

²In the one-dimensional setting considered here, ε reduces to the axial component ε_{xx} ; the generalization to tensorial strain is straightforward through the appropriate invariants.

reference state $\varepsilon = 0$:

$$\sigma(\varepsilon) \approx \sigma_0 + \left. \frac{d\sigma}{d\varepsilon} \right|_0 \varepsilon + \frac{1}{2} \left. \frac{d^2\sigma}{d\varepsilon^2} \right|_0 \varepsilon^2 + \mathcal{O}(\varepsilon^3), \quad (16)$$

where $\sigma_0 = \sigma(0)$ is the reference (unstrained) conductivity.

Factoring out σ_0 yields the normalized form

$$\sigma(\varepsilon) = \sigma_0 (1 + \pi \varepsilon + \alpha \varepsilon^2 + \dots), \quad (17)$$

with the dimensionless coefficients

$$\pi = \frac{1}{\sigma_0} \left. \frac{d\sigma}{d\varepsilon} \right|_0, \quad \alpha = \frac{1}{2\sigma_0} \left. \frac{d^2\sigma}{d\varepsilon^2} \right|_0. \quad (18)$$

Here π is the *linear piezoresistive coefficient*, characterizing the first-order sensitivity of the conductivity to strain, while α captures higher-order nonlinear effects relevant at larger strain amplitudes. The expansion (17) is not geometric in nature; rather, it provides a local approximation of how the *internal electronic transport properties* of the material respond to deformation.

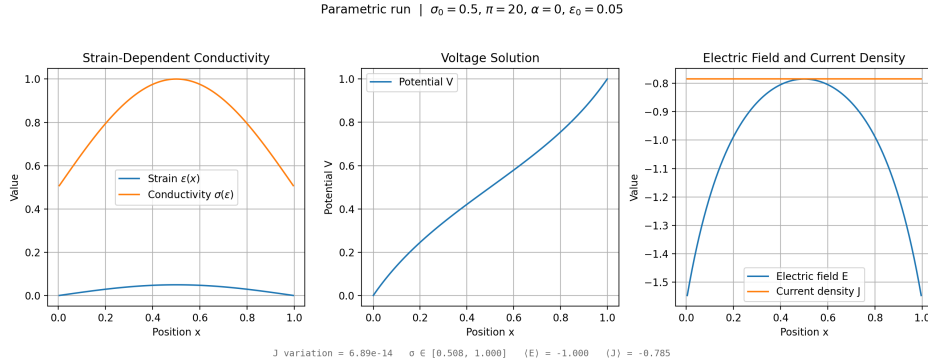


Figure 4: Normalized conductivity $\sigma(\varepsilon)/\sigma_0$ as a function of strain ε for representative values of the linear coefficient π and the nonlinear coefficient α . For small $|\varepsilon|$ the response is dominated by the linear term; the quadratic contribution becomes visible at larger strain amplitudes.

Physically, mechanical strain modifies the crystal lattice, which in turn alters the electronic band structure, the carrier effective masses, and the dominant scattering mechanisms. Within the Drude picture, the conductivity reads

$$\sigma = \frac{n e^2 \tau}{m^*}, \quad (19)$$

where n is the carrier density, e the elementary charge, τ the momentum relaxation time, and m^* the effective mass. Strain therefore enters σ through all three microscopic quantities n , τ , and m^* simultaneously. Consequently, the Taylor expansion (17) should be understood as an effective continuum-level description that subsumes these microscopic contributions into the macroscopic coefficients π and α , without requiring an explicit microscopic derivation.

The parametric study based on (17) reveals a clear separation between uncoupled and coupled regimes. For $\pi = \alpha = 0$ the conductivity reduces to the constant baseline $\sigma = \sigma_0$, recovering a linear potential profile $V(x)$, a constant electric field, and a uniform current density; strain is present but does not influence transport. A non-zero linear coefficient π activates first-order piezoresistive coupling: the conductivity now responds directly to the strain field, the electric field redistributes

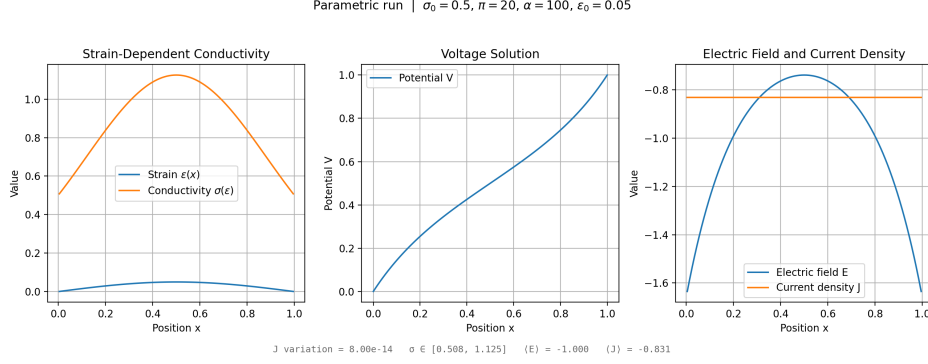


Figure 5: Finite element results for the one-dimensional coupled problem with strain-dependent conductivity (17). A prescribed strain field $\varepsilon(x) = \varepsilon_0 \sin(\pi x)$ is used to illustrate the effect of the coefficients π and α on the potential, electric field, and current density. The current density remains constant up to machine precision, confirming current conservation.

accordingly, and the potential profile becomes nonlinear, while the current density remains spatially constant in accordance with charge conservation,

$$\frac{dJ}{dx} = 0. \quad (20)$$

The sign and magnitude of π control whether the conductivity locally increases or decreases under deformation, and thereby the sensitivity of the electrical response. The quadratic term $\alpha\varepsilon^2$ introduces additional curvature in $\sigma(\varepsilon)$ and is typically subdominant for small strains, but becomes appreciable at larger amplitudes. In all cases the admissibility condition $\sigma(\varepsilon) > 0$ must be respected to preserve physical meaning.

Across the tested configurations, the numerical results confirm that the current density is conserved to machine precision, validating the finite element implementation. The simulations show that the system maintains equilibrium by compensating local variations in conductivity with corresponding adjustments in the electric field, consistent with the constitutive relation $J = \sigma E$. Representative parameter ranges for stable exploration are $\sigma_0 \in [0.1, 2.0]$, $\pi \in [5, 20]$, $\alpha \in [0, 100]$, and strain amplitudes $\varepsilon_0 \in [10^{-3}, 10^{-2}]$. Within this regime, the model captures both linear and nonlinear piezoresistive behavior while remaining numerically robust and physically interpretable.

16 Physics-Informed Conductivity from First Principles

The parametric study of Section 15 treated the coefficients π and α in

$$\sigma(\varepsilon) = \sigma_0(1 + \pi\varepsilon + \alpha\varepsilon^2 + \dots) \quad (21)$$

as free phenomenological parameters. This section removes that freedom in two steps. First, deformation-potential theory fixes the *form* of $\sigma(\varepsilon)$ and expresses π, α in terms of a single material coupling, the hydrostatic deformation potential ($a_c + a_v$). Second, that coupling—together with the shear deformation potentials—is computed directly from density-functional theory (DFT), so the constitutive law is fixed entirely from first principles rather than fitted. The overall pipeline is

$$\varepsilon \longrightarrow H_s \longrightarrow \Delta E_g(\varepsilon) \longrightarrow n(\varepsilon) \longrightarrow \sigma(\varepsilon) \longrightarrow \nabla \cdot (\sigma \nabla V) = 0, \quad (22)$$

with each arrow developed below, following the treatment in Hamaguchi [3].

16.1 Strain Hamiltonian and band-gap shift

When a crystal is uniformly strained, the periodic potential seen by the electrons changes, perturbing the Bloch Hamiltonian. To first order in strain this perturbation is called the strain Hamiltonian H_s ; it couples the electronic degrees of freedom linearly to the strain tensor ε_{ij} through the deformation-potential operator D_{ij} [3]:

$$H = H_0 + H_s, \quad H_s = \sum_{i,j} D_{ij} \varepsilon_{ij}. \quad (23)$$

Here H_0 is the unstrained Bloch Hamiltonian and D_{ij} encodes how each strain component modifies the crystal potential; its explicit form involves derivatives of the ionic pseudopotential but need not be evaluated directly—only its band-state matrix elements appear in what follows. To first order, the shift of a non-degenerate band edge $|n\rangle$ follows from perturbation theory,

$$\Delta E_n = \langle n | H_s | n \rangle = \sum_{i,j} \langle n | D_{ij} | n \rangle \varepsilon_{ij}. \quad (24)$$

Separating the strain into a hydrostatic part,

$$\varepsilon_h = \varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz} = \text{Tr } \varepsilon, \quad (25)$$

and traceless shear components, the hydrostatic part shifts the band edges rigidly while the shear part lifts degeneracies. The shear-induced splitting, parameterized by the deformation potentials b (equivalently the uniaxial potential Ξ_u) and d in the Picus–Bir framework, is anisotropic and underlies the dominant piezoresistive response of doped silicon; it is quantified in Section 16.6 and its route into transport is discussed in Section 16.8. The hydrostatic shift of the gap is

$$\Delta E_g(\varepsilon_h) \approx (a_c + a_v) \varepsilon_h, \quad (26)$$

where a_c and a_v are the hydrostatic deformation potentials of the conduction and valence band edges [3].

16.2 From band shift to carrier density

In the non-degenerate Maxwell–Boltzmann limit the conduction-band electron concentration is

$$n = N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right), \quad (27)$$

and for an intrinsic semiconductor charge neutrality fixes the Fermi level near mid-gap, giving

$$n_i = \sqrt{N_c N_v} \exp\left(-\frac{E_g}{2k_B T}\right), \quad (28)$$

so that the strain-induced gap shift (26) produces a multiplicative change in n_i , with the factor 1/2 inherited from the square root.

16.3 Strain-dependent conductivity

Combining the Drude form $\sigma = ne^2\tau/m^*$ with the intrinsic carrier density (28) and the linear gap shift (26), and assuming the mobility prefactor τ/m^* varies weakly with strain,

$$\sigma(\varepsilon) \approx \sigma_0 \exp\left(-\frac{\Delta E_g(\varepsilon)}{2k_B T}\right) = \sigma_0 \exp\left(-\frac{(a_c + a_v)\varepsilon_h}{2k_B T}\right), \quad (29)$$

where σ_0 absorbs the unstrained prefactors. Introducing the dimensionless coupling

$$\kappa \equiv \frac{a_c + a_v}{2k_B T}, \quad (30)$$

expanding the exponential for $|\kappa\varepsilon_h| \ll 1$, and matching term by term with (21) gives

$$\pi = -\kappa = -\frac{a_c + a_v}{2k_B T}, \quad \alpha = \frac{1}{2}\kappa^2 = \frac{1}{2}\left(\frac{a_c + a_v}{2k_B T}\right)^2. \quad (31)$$

Equation (31) is the key link: the phenomenological coefficients acquire a microscopic meaning, their sign and relative magnitude fixed by a single material parameter. For typical semiconductors with $(a_c + a_v) > 0$, the linear coefficient π is negative—tensile hydrostatic strain widens the gap and lowers the intrinsic conductivity—and $\alpha = \frac{1}{2}\pi^2$, so the two coefficients are not independent.

16.4 Sensitivity to the coupling strength

Before fixing $(a_c + a_v)$ from DFT, it is useful to map how the response depends on it. Table 1 evaluates (31) at $T = 300$ K ($k_B T = 0.0259$ eV) for three representative couplings. Even a modest coupling of 1 eV gives $|\pi| \approx 19$, at the upper end of the phenomenological range $\pi \in [5, 20]$ used in Section 15; the first-principles value obtained below (Section 16.7) falls between the moderate and strong cases.

case	$a_c + a_v$ [eV]	ε_0	π	α
weak	0.3	5×10^{-3}	-5.80	1.68×10^1
moderate	1.0	5×10^{-3}	-19.34	1.87×10^2
strong	3.0	3×10^{-3}	-58.02	1.68×10^3
this work (DFT)	1.83	$\leq 10^{-2}$	-35.4	6.3×10^2

Table 1: Coefficients π and α from (31) at $T = 300$ K. The first three rows are a sensitivity sweep over representative couplings; the last row uses the first-principles value $a_c + a_v = 1.83$ eV determined in Section 16.6.

The exponential law (29) and its Taylor truncation (21) are compared in Figure 6. For weak and moderate coupling the two are indistinguishable across $|\varepsilon| \leq 0.02$; for strong coupling the Taylor form begins to miss the tensile/compressive asymmetry of the exponential.

Figure 6: Normalized conductivity $\sigma(\varepsilon)/\sigma_0$ versus strain for three couplings at $T = 300$ K. Solid: exponential law (29); dashed: Taylor approximation (21) with the coefficients (31).

16.5 Coupled finite-element solution

The one-dimensional conductivity equation (13) is solved with the physics-informed law (29) for a prescribed strain profile $\varepsilon(x) = \varepsilon_0 \sin(\pi x)$ on $[0, 1]$ with $V(0) = 0, V(1) = 1$. The mechanical assembly, linear solve, and post-processing are unchanged from Section 15; only the local material law is replaced.

Figure 7 shows the moderate case ($a_c + a_v = 1 \text{ eV}$, $\varepsilon_0 = 5 \times 10^{-3}$): the strain modulates the conductivity, the potential bows away from the linear baseline, the electric field develops a compensating spatial variation, and the current density remains constant to machine precision—the discrete statement of $\nabla \cdot J = 0$ and the principal consistency check. The exponential and Taylor laws are numerically indistinguishable at this amplitude, consistent with (31). Figure 8 shows that the departure of $V(x)$ and $E(x)$ from the uncoupled baseline grows with $(a_c + a_v)$.

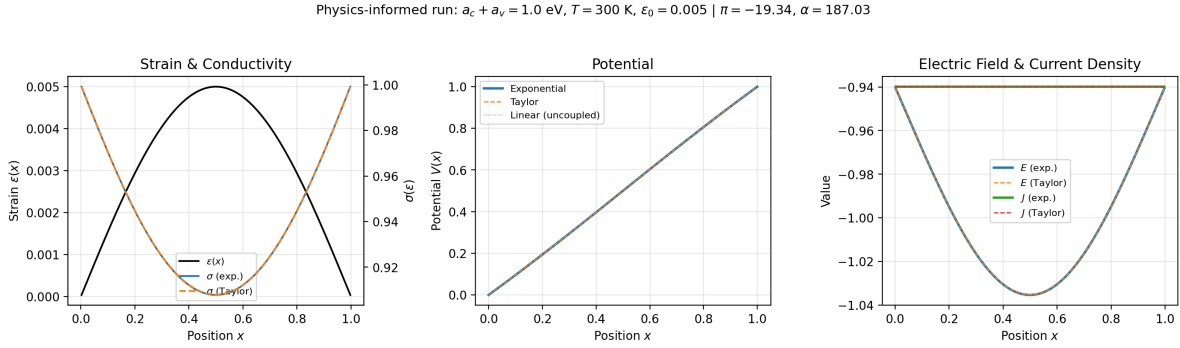


Figure 7: Finite-element solution of the coupled problem with the physics-informed law (29) for $a_c + a_v = 1 \text{ eV}$ at $T = 300 \text{ K}$, $\varepsilon_0 = 5 \times 10^{-3}$. Left: strain and conductivity; middle: potential; right: electric field and current density. J is constant to machine precision while E varies to compensate $\sigma(x)$.

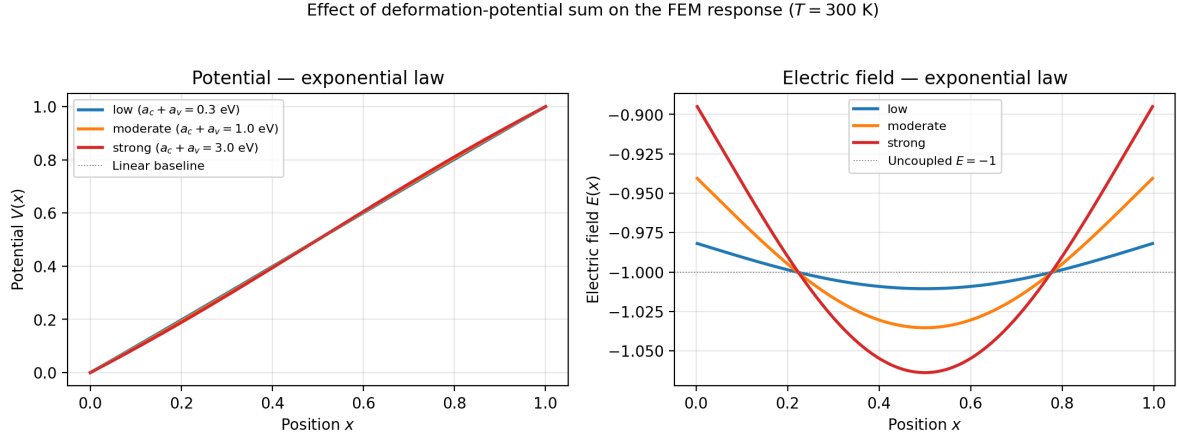


Figure 8: Effect of the deformation-potential sum on the FEM response at $T = 300 \text{ K}$. Left: potential profiles for the three cases of Table 1; right: electric field. The deviation from the uncoupled value $E = -1$ grows with the coupling strength.

16.6 First-principles deformation potentials

The coupling ($a_c + a_v$) used above is only one of three independent deformation potentials of cubic silicon. The other two—the uniaxial potential Ξ_u and the shear potential d —do not enter the scalar law (29), but they govern the dominant, anisotropic part of silicon’s piezoresistance through valley repopulation and valence-band splitting (Section 16.8). All three are computed here from the band structure: the hydrostatic potential to fix the constitutive law, and the shear potentials to supply the inputs the full transport treatment requires. The physical chain is exactly the pipeline (22): applied strain moves the atoms, which changes the potential the electrons feel, which shifts and splits the bands; the slopes of those shifts with respect to strain are the deformation potentials. Cubic symmetry fixes how many are independent and which strain isolates each one.

One strain mode per deformation potential. A general strain of silicon separates into three symmetry-distinct modes, each isolating one deformation potential (Table 2). This is what makes the calculation efficient: three symmetry-pure strains suffice, rather than a scan over arbitrary deformations.

Strain mode	Effect on the bands	Probes
Hydrostatic (volume)	gap shifts, no splitting	$a = a_c + a_v$
Tetragonal (one axis)	$\langle 100 \rangle$ valleys split	Ξ_u
Trigonal shear (angles)	valence top at Γ splits	d

Table 2: The three symmetry-pure strain modes and the deformation potential each isolates. Hydrostatic strain keeps all directions equivalent (no splitting); the tetragonal and shear modes break symmetry and lift specific degeneracies.

Computational method. Calculations use VASP with PAW PBE potentials, a plane-wave cut-off of 320 eV, and the relaxed primitive cell of silicon as the $\varepsilon = 0$ reference. Crystal symmetry is disabled (`ISYM=0`) so that a small symmetry-lowering strain is not folded back to the cubic structure, which would erase the splitting being measured. Each strained structure is run in stages: a self-consistent calculation on a uniform $\mathbf{k}$ -mesh converges the charge density, then a non-self-consistent calculation produces the bands along a high-symmetry path. For the trigonal-shear runs an ionic relaxation at fixed cell shape precedes these steps, because the two atoms of the diamond basis displace relative to one another under shear (the internal-strain or Kleinman effect); the hydrostatic and tetragonal modes need no such relaxation. Strain amplitudes of $\pm 0.2, 0.5, 1.0\%$ are applied, and each deformation potential is obtained as the slope of a band feature versus strain. The equilibrium PBE band structure is shown in Figure 9; PBE underestimates the absolute gap ($E_g(0) \approx 0.62$ eV versus the experimental 1.12 eV), but the deformation potentials are *slopes* and are far less sensitive to this error than the absolute gap.

Review Later 1 *However one can use corrections for example in [7].*

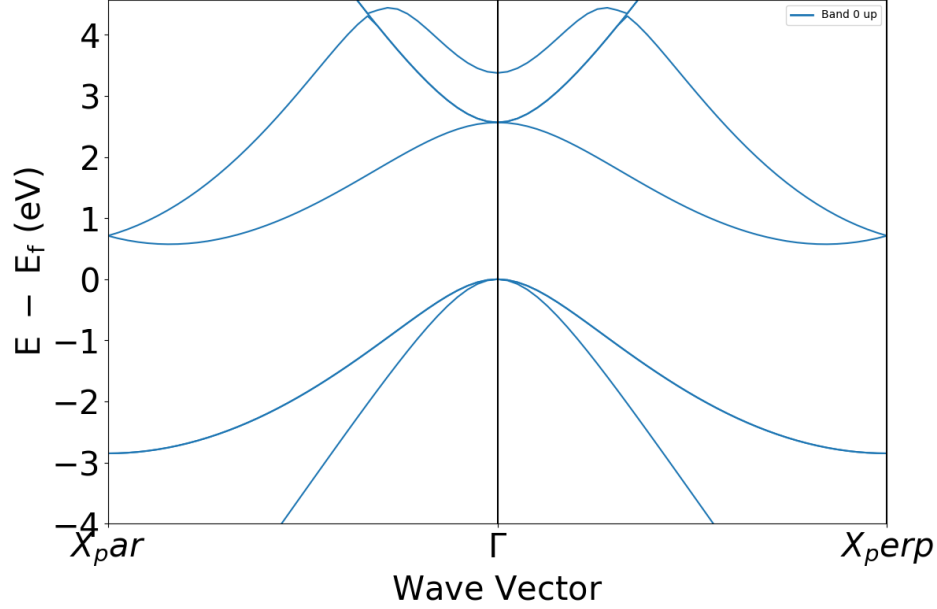


Figure 9: Equilibrium band structure of silicon (VASP, PBE) along the dual-X path $X_{\parallel}$ - Γ - $X_{\perp}$, energies referenced to the valence-band maximum. At zero strain the two X points are degenerate; the valence maximum sits at Γ and the conduction minima near X , reproducing the indirect gap.

Extraction and results. The extraction relations follow from the band features in Table 2:

$$a = a_c + a_v = \frac{dE_g}{d \text{Tr } \varepsilon}, \quad (32)$$

$$\Xi_u = \frac{d}{d\varepsilon} [E_c(X_{\parallel}) - E_c(X_{\perp})], \quad (33)$$

$$|d| = \frac{\Delta E_{\Gamma_{25'}}}{3\sqrt{3}\varepsilon_{xy}}. \quad (34)$$

The band structures for the three modes at $\varepsilon = -1\%, 0, +1\%$ are shown in Figure 10, and the linear fits behind each potential in Figure 11. The resulting values are collected in Table 3.

Deformation potential	This work	Literature (Si)	Source
$a = a_c + a_v$	+1.83 eV	$\sim +1.5$	[8]
Ξ_u	+8.89 eV	9.0 ± 2	[8, 4]
$ d $	5.46 eV	~ 5.3 ($d < 0$)	[8, 5]
$dE_g/dP = -a/B$	-18.7 meV/GPa	~ -15	[8]

Table 3: Deformation potentials of silicon from the three strain sweeps, compared with literature. The gap potential a carries its sign because it sets the sign of π via (31); Ξ_u is a positive splitting coefficient; d is reported as a magnitude, the standard convention being $d < 0$. The pressure coefficient $dE_g/dP = -a/B$ (bulk modulus $B \approx 98$ GPa) is a convention-independent comparison with experiment.

Each sweep reproduces the behaviour symmetry requires, which validates the calculation independently of the fitted numbers. Hydrostatic strain shifts the gap monotonically with no splitting;

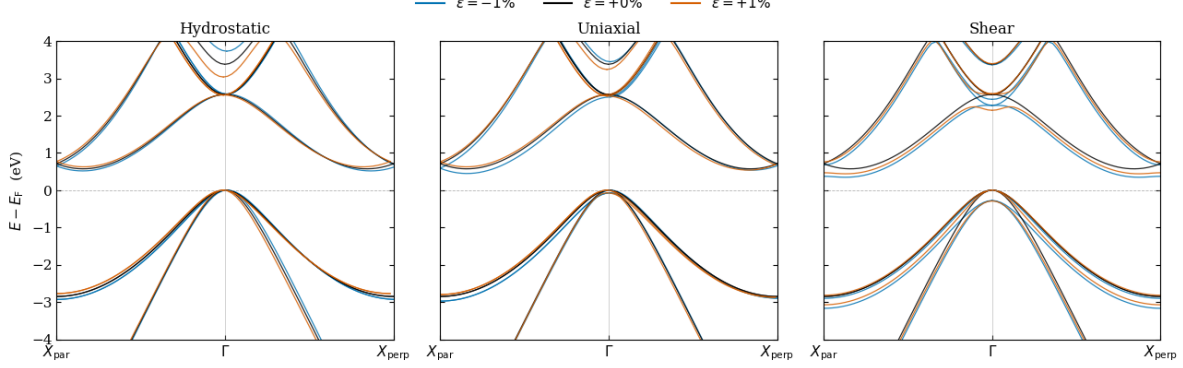


Figure 10: Band structure of silicon at $\varepsilon = -1\%, 0, +1\%$ for the three strain modes. Hydrostatic strain shifts the gap with no splitting; tetragonal strain splits the conduction valleys ($X_{\parallel}$ vs. $X_{\perp}$); trigonal shear splits the valence-band maximum at Γ .

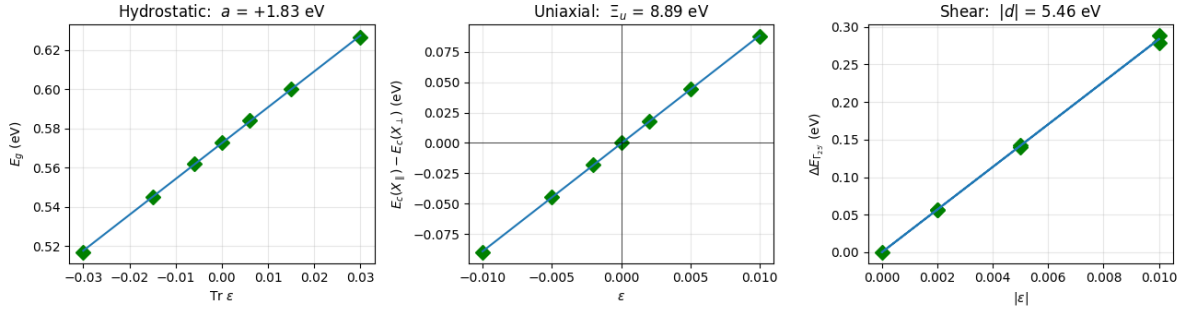


Figure 11: Deformation potentials as slopes: gap versus hydrostatic strain (a); valley splitting versus tetragonal strain (Ξ_u), passing through zero at $\varepsilon = 0$; valence splitting versus shear magnitude ($|d|$). The slopes give the values of Table 3.

tetragonal strain produces a valley splitting that vanishes at $\varepsilon = 0$ and reverses sign with the strain (the transverse valleys forming the conduction minimum under tension, the strain-axis valleys under compression); trigonal shear splits the $\Gamma_{25'}$ valence triplet symmetrically in $\pm\varepsilon$, the signature of a pure traceless shear. The negative pressure coefficient, close to the measured value, confirms the full chain from atomic structure to band gap.

16.7 First-principles π and α

Inserting the computed $a = a_c + a_v = 1.83 \text{ eV}$ into (30) and (31) at $T = 300 \text{ K}$ gives

$$\kappa = \frac{a_c + a_v}{2k_B T} = 35.4, \quad \pi = -35.4, \quad \alpha = \frac{1}{2}\kappa^2 = 626. \quad (35)$$

This is the last row of Table 1: silicon sits between the moderate and strong cases, the coupling is sizeable, and the negative sign confirms that tensile hydrostatic strain lowers the intrinsic conductivity. The representative parameters of Section 15 are thereby replaced by a single computed value, and the hydrostatic part of $\sigma(\varepsilon)$ in (29) is fixed entirely from first principles. Figure 12 shows the resulting finite-element solution; the potential deviates visibly from the uncoupled baseline, confirming that $a = 1.83 \text{ eV}$ produces a physically meaningful piezoresistive response.

First-principles run: $a_c + a_v = 1.83$ eV (DFT), $T = 300$ K, $\varepsilon_0 = 0.005$ | $\pi = -35.4$, $\alpha = 626$

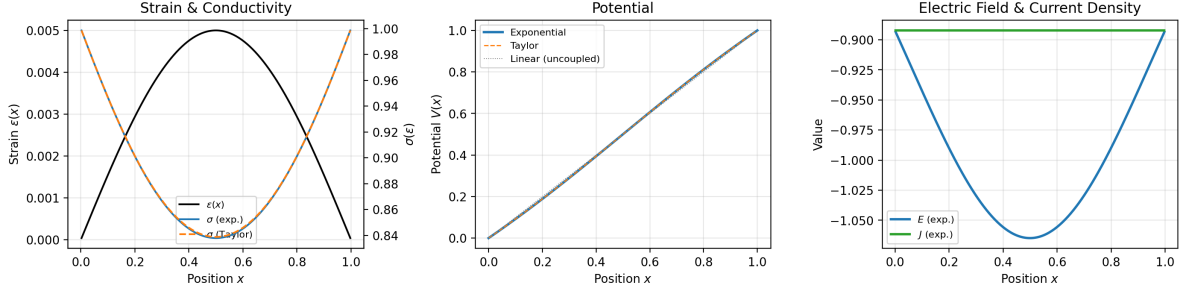


Figure 12: Finite-element solution with the physics-informed law (29) for $a_c + a_v = 1.83$ eV (DFT, $\pi = -35.4$, $\alpha = 626$) at $T = 300$ K, $\varepsilon_0 = 5 \times 10^{-3}$. Left: strain and conductivity; middle: potential; right: electric field and current density. J is constant to machine precision while E compensates $\sigma(x)$, completing the first-principles pipeline (22).

16.8 Outlook: the transport step and the role of shear strain

The constitutive law developed above is complete and self-contained, but it captures only one of the two channels through which strain alters conductivity, and not the dominant one for the devices of interest.

Two channels, one implemented. Through $\sigma = ne\mu$, strain can act on the carrier density n or on the mobility μ . The exponential law (29) is the *density* channel: hydrostatic strain shifts the gap, the gap shifts n through Maxwell–Boltzmann statistics, and only the hydrostatic potential $a = a_c + a_v$ enters. This channel is isotropic by construction—it depends on ε only through the scalar $\text{Tr } \varepsilon$ —and is sub-dominant in doped silicon, where $n \approx N_D$ is pinned by the dopants and responds only weakly to a gap shift.

Why the shear potentials were computed. The large, technologically exploited piezoresistance of silicon, first measured by Smith [6], comes from the *mobility* channel and is anisotropic. Uniaxial $\langle 100 \rangle$ strain splits the six conduction valleys (coefficient Ξ_u); electrons repopulate toward the lowered valleys, changing the direction-averaged conductivity mass and hence μ . This is the origin of the large π_{11} of n -type silicon. Trigonal shear splits the valence-band maximum (coefficient d), redistributing holes among bands of different curvature, which dominates π_{44} of p -type silicon. The potentials Ξ_u and d in Table 3 are therefore not auxiliary: they parameterize the principal piezoresistive effect. They are absent from $\sigma(\varepsilon)$ here only because the carrier-density argument has no place for them—valley repopulation changes mobility, not carrier number.

Where the Boltzmann transport equation enters. Routing Ξ_u and d into transport requires the conductivity in its full Boltzmann tensor form,

$$\sigma_{ij} = \frac{e^2}{4\pi^3} \sum_n \int v_i^n(\mathbf{k}) v_j^n(\mathbf{k}) \tau_n(\mathbf{k}) \left(-\frac{\partial f_0}{\partial E} \right) d\mathbf{k}, \quad (36)$$

rather than the scalar $\sigma = ne\mu$. Strain enters (36) through the shifted band energies $\delta E_n = \Xi_{ij}\varepsilon_{ij}$, which modify the group velocities $v_i^n = \hbar^{-1} \partial E_n / \partial k_i$, the valley occupations through $-\partial f_0 / \partial E$, and the intervalley relaxation times τ_n . The result is a genuine tensor $\sigma_{ij}(\varepsilon)$ carrying the anisotropy

that the scalar law cannot represent. Equation (36) requires two ingredients the present work does not yet supply: the strained band structure on a dense $\mathbf{k}$ -mesh, available in principle from the same VASP runs used here, and the relaxation times $\tau_n(\mathbf{k})$, which demand a separate electron-phonon calculation (density-functional perturbation theory coupled to a Boltzmann-transport solver). The deformation potentials of Section 16.6 are exactly the band-structure inputs this step consumes; the relaxation times are the remaining piece.

Further approximations. Two narrower assumptions accompany the density channel. The intrinsic carrier density (28) overstates the gap contribution in doped devices, where the mobility channel above dominates; and the Maxwell-Boltzmann form (27) fails at high doping or low temperature, where Fermi-Dirac statistics are required. Neither alters the structure of the pipeline (22) or the finite-element infrastructure of Section 15: each refinement replaces the local material law while leaving the assembly and solver untouched.

16.9 Uniaxial Channel: Valley Repopulation

In hydrostatic channel we have developed through 23 (the trace part of strain tensor), the shift of bands calculations. We have also identified the anisotropic splitting of the bands in traceless part of the tensor.

The route to transport for this part includes the electron mobility repopulation since the anisotropic distribution of the valleys. This leads to study the redistribution of energy valleys, as an effect of applied uniaxial strain.

Symmetry of the six-valley splitting. The uniaxial strain breaks the rotational symmetry(...). This leads to an anisotropic repopulation of the momenta along $\langle 100 \rangle$, for Si and along $\langle 111 \rangle$ for Ge /citehamaguchi.

Repopulation and the conductivity tensor.

Extraction of $\pi_{11} - \pi_{12}$.

Role of the effective masses.

17 Expected Outcome

The expected outcome of this study is the development of a computational framework for analyzing strain-dependent electrical conduction in deformable materials. The framework combines continuum mechanics and electrical transport into a unified numerical model, enabling the systematic investigation of electromechanical coupling.

At the modeling level, the work provides a consistent formulation of conductivity as a strain-dependent quantity, together with a numerical implementation for solving the resulting coupled system. This allows for the analysis of how mechanical deformation influences electrical potential distribution, current density, and overall system sensitivity.

From a computational perspective, the study delivers a flexible pipeline that can accommodate different forms of the strain-conductivity relation, ranging from phenomenological descriptions to physically informed or data-driven models. This makes the framework adaptable to different materials and configurations.

Beyond the specific application, the results contribute to the understanding of nonlinear multiphysics systems involving coupling between deformation and transport phenomena. Potential applications include piezoresistive sensing, structural health monitoring, and functional materials design.

The developed framework also provides a foundation for future extensions toward multiscale modeling and more detailed transport descriptions.

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